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In [1]: # Binomial Distribution
from scipy.stats import binom

n = 5
k = 3
p = 0.5

prob = binom.pmf(k, n, p)
print(prob)    # 5/ 16
```

0.31249999999999994

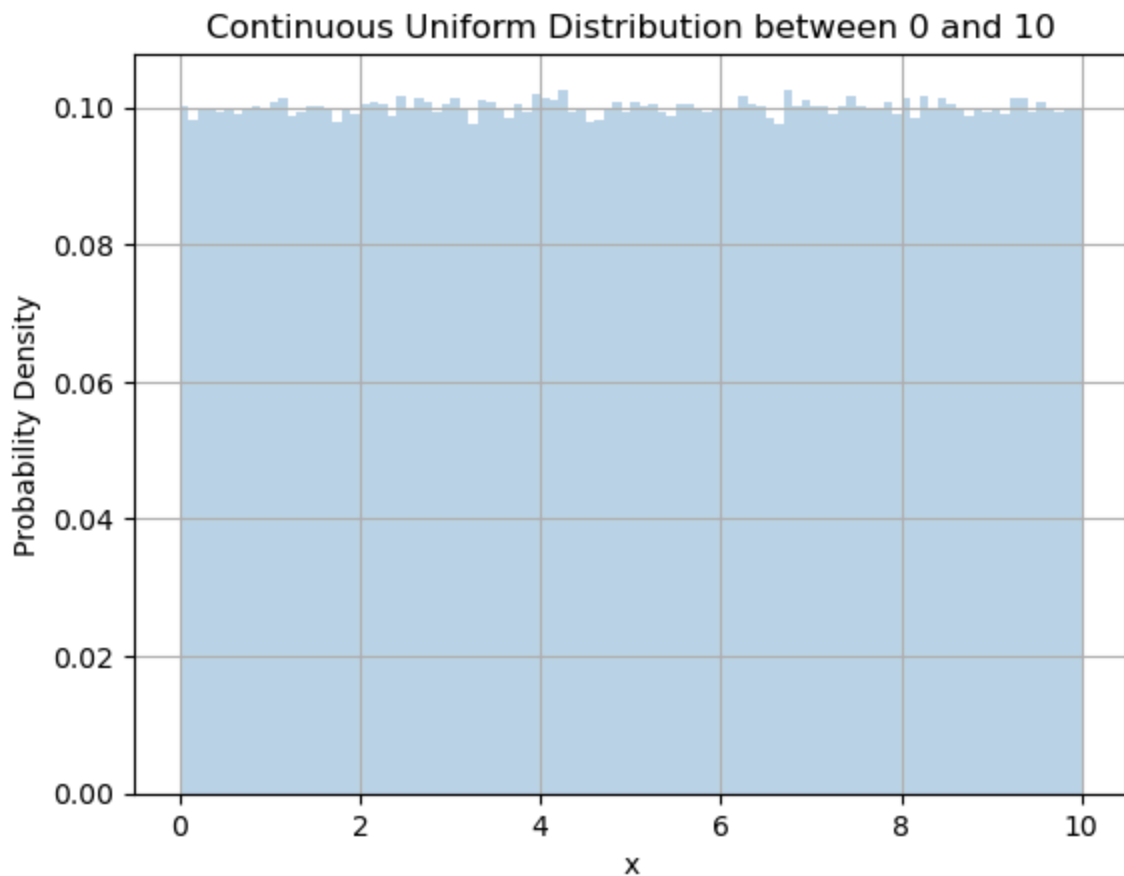
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In [4]: # Uniform Distribution
import numpy as np
import matplotlib.pyplot as plt

# values from 0 to 10
values = np.random.uniform(0, 10, 10_00_000)

# plotting a histogram - becomes more uniform with more values
plt.hist(values, bins=100, density=True, alpha=0.3)

plt.title("Continuous Uniform Distribution between 0 and 10")
plt.xlabel("x")
plt.ylabel("Probability Density")
plt.grid(True)

plt.show()
```



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In [5]: # Normal Distribution
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# Parameters
mu = 70      # mean
sigma = 10   # standard deviation

# X-axis values
x = np.linspace(30, 110, 1000)

# Normal PDF
y = norm.pdf(x, mu, sigma)

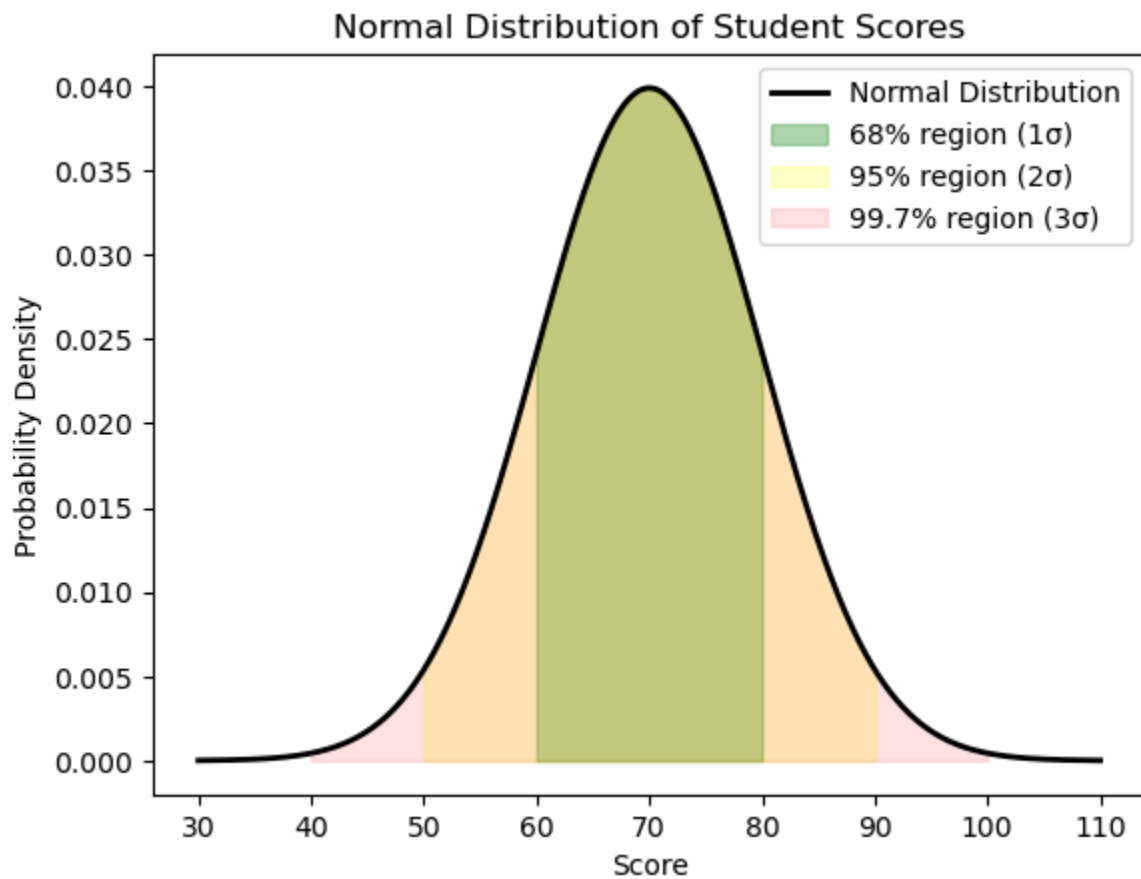
# Plot the bell curve
plt.plot(x, y, color='black', linewidth=2, label='Normal Distribution')

# Shade 68% region ( $\mu \pm 1\sigma$ )
x1 = np.linspace(mu - sigma, mu + sigma, 1000)
plt.fill_between(x1, norm.pdf(x1, mu, sigma), color='green', alpha=0.3, label=

# Shade 95% region ( $\mu \pm 2\sigma$ )
x2 = np.linspace(mu - 2*sigma, mu + 2*sigma, 1000)
plt.fill_between(x2, norm.pdf(x2, mu, sigma), color='yellow', alpha=0.2, label=
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# Shade 99.7% region ( $\mu \pm 3\sigma$ )
x3 = np.linspace(mu - 3*sigma, mu + 3*sigma, 1000)
plt.fill_between(x3, norm.pdf(x3, mu, sigma), color='red', alpha=0.1, label='9

# Labels and title
plt.title("Normal Distribution of Student Scores")
plt.xlabel("Score")
plt.ylabel("Probability Density")
plt.legend()
plt.show()
```



In []: